

TASK 4 SUBMISSION

Competitor Demand Prediction Using Job Postings

Skill Extraction and Feature Engineering

Extracted skills and feature tables for NVIDIA

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Company specialism	NVIDIA
Task	Task 4 of 10 - Skill Extraction and Feature Engineering
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Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Measure	Starting point	Final
Canonical skills in the taxonomy	59	93
Postings matching at least one skill	370 (21.2%)	1,256 (72.0%)
Mean skills per posting	0.24	1.03
Total skill mentions	419	1,793
Distinct skills matched	27	59

The finding this task turned on

The shared taxonomy covered only 21.2% of NVIDIA's postings, because it was written for job descriptions and my source carries only titles. A description names the tool - Verilog, Synopsys, PrimeTime. A title names the specialisation - Senior Physical Design Engineer. 32 of the original 59 skills never matched once. Extending the taxonomy to the vocabulary titles actually use raised coverage to 72.0%.

1 The problem with the taxonomy

The starter taxonomy lists tools: AWS, Docker, Spark, TensorFlow, Tableau. Those words belong in a job description. A job title names the specialisation, not the tool.

Measured against 1,745 NVIDIA titles, **32 of the 59 taxonomy skills never matched once**: SQL, Java, Scala, TensorFlow, scikit-learn, Spark, Kafka, Airflow, dbt, AWS, Azure, GCP, Docker, Terraform, MLflow, Tableau, Power BI, A/B Testing, Statistics, RAG, RLHF and others. That is not a statement about NVIDIA's technology. It is a statement about what fits in a job title.

1.1 What I added

34 canonical skills, in a new `silicon_design` category plus additions to existing ones.

Area	Added
<code>silicon_design</code> (new)	Physical Design, RTL Design, Silicon Verification, SoC Design, Analog and Mixed Signal, Signal and Power Integrity, Design for Test, Design Automation, CPU Architecture, GPU Architecture
Systems	Firmware, Kernel and Drivers, Compilers, Networking, DPU and Offload, Datacenter Systems, Storage Systems, Performance Engineering
Applied AI	Generative AI, Agentic AI, Inference and Serving, Autonomous Vehicles, Digital Twin and Simulation, Graphics and Rendering, Healthcare AI, Speech and Audio
Practice and go-to-market	Cloud Infrastructure, Site Reliability, Cybersecurity, Test Automation, Solutions Architecture, Developer Relations, Technical Program Management, Business Development

Role words are deliberately **not** skills. Engineer, manager, architect and director are `role_function`, derived in Task 3. Letting them into the skill space would make every posting look identical.

All additions went into `shared/taxonomy/skills.yaml` with a dated comment block explaining the reasoning, per the governance rule in `docs/skill-taxonomy.md`. **They need team ratification**, and that is the main thing I am bringing to the sprint meeting.

2 Method

Dictionary matching of canonical names and aliases against the cleaned title text from Task 3, with three rules that exist because the input is a title:

- Longest alias wins.** *gpu architecture* must not also register as a bare *gpu*, and *computer vision* must not register as *vision*. Aliases are sorted by length and a matched span is consumed, so no shorter term can claim it.
- Blocked single letters.** The taxonomy carries `title_match_blocklist`: `[r, c]`. Both were measured, not assumed: *r* matched **R&D** in 9 postings once punctuation was stripped, and *c* matched the *c* inside *c++* in 11, double-counting *C* and *C++*.
- No inference.** A title reading *Senior Software Engineer* and nothing else gets no skill. Guessing that a software engineer at NVIDIA probably knows C++ would manufacture data.

Libraries: `pandas`, `pyyaml`, and `re` from the standard library. The taxonomy is the single source of truth; the script contains no skill list of its own.

3 Validation

20 postings were checked by hand. That found one false positive and one large miss.

False positive: Design Automation tagged as Test Automation

Senior Design Automation Engineer, Applied AI was tagged Test Automation, because "automation" was an alias of it. Design Automation is EDA tooling, a different thing entirely. The bare alias was removed and Design Automation added as its own silicon skill. Test Automation fell from 36 matches to 14, which is the honest number.

Large miss: one missing word form cost 192 postings

Senior Solutions Architect, AI Infrastructure matched nothing, because the taxonomy had "solutions architecture" but titles say "solutions architect". That single word form covers 11% of the corpus, and once fixed it became the largest skill in the dataset.

Rare misses were left as candidates rather than fixed: silicon photonics (2 postings), NeMo (3), OpenBMC (2), linear algebra (1). Adding a canonical skill for a term seen twice would inflate the taxonomy for no analytical gain.

Coverage across the four passes: 21.2% → 63.5% after the extension → 67.2% after adding Cloud Infrastructure and Site Reliability → **72.0%** after the validation fixes.

4 A bias I had to correct

The emerging-skill flag compares the first half of the window with the second. Run naively it made almost everything look like it was declining, and the reason is in the Task 2 data: **posting_date is a first-seen proxy, so every posting already open at the first observation is stamped with that date** - 395 of 1,745 postings, 23% of the corpus, all landing in the first half.

The trend basis therefore excludes the first observation entirely, comparing 641 postings against 709. Nothing else in the task uses that exclusion. Before the fix, Datacenter Systems appeared to fall from 82 to 36; after it, from 49 to 36, which is a real but far milder decline.

5 Results

5.1 Where NVIDIA's hiring sits

hardware_systems	629	
research_domains	334	
product_process	269	
silicon_design	256	
cloud_infra	147	
mlops_devops	119	
languages	25	
ml_frameworks	14	

Hardware and silicon together account for 885 of 1,793 mentions, just under half. ML frameworks, the thing most people associate with NVIDIA, account for 14. **NVIDIA is hiring the people who build the machine, not the people who use it.**

5.2 Top 12 skills

#	Skill	Category	Postings	Share
1	Solutions Architecture	product_process	192	11.0%
2	Networking	hardware_systems	142	8.1%
3	Cloud Infrastructure	cloud_infra	133	7.6%
4	Datacenter Systems	hardware_systems	118	6.8%
5	Performance Engineering	hardware_systems	113	6.5%
6	Deep Learning	research_domains	81	4.6%
7	Silicon Verification	silicon_design	79	4.5%
8	SoC Design	silicon_design	61	3.5%
9	Developer Relations	product_process	57	3.3%
10	Agentic AI	research_domains	56	3.2%
11	Compilers	hardware_systems	55	3.2%
12	Cybersecurity	mlops_devops	55	3.2%

5.3 Emerging skills

12 skills flagged, meaning their rate in the second half exceeded 1.5 times the first half on at least 5 postings. The eight strongest:

Skill	First half	Second half	Skill	First half	Second half
Inference and Serving	3	14	HPC	14	24
Compilers	11	25	LLMs	4	8
Agentic AI	17	31	Design Automation	3	9
AI Safety	0	7	Physical Design	6	13

AI Safety going from zero to seven is the cleanest signal in the set, because it cannot be a sampling artefact of the kind described in section 4.

5.4 Seniority concentration

Most senior	Senior+	Least senior	Senior+
Storage Systems	94%	Business Development	21%
C++	91%	Robotics	46%
Analog and Mixed Signal	90%	Developer Relations	47%
Signal and Power Integrity	90%	SoC Design	54%

The analog and signal-integrity roles being almost entirely senior is consistent with a scarce talent pool. Nothing in this data proves that, and it is stated as an observation, not a finding.

6 Limitations

#	Limitation
1	28% of postings carry no skill. 489 titles name a role and nothing else. They are not skill-free jobs; the title is simply silent.
2	One skill per posting on average. A description-based extraction would yield 15 to 25. Any comparison against a member who has descriptions is a comparison of sources, not of companies.
3	Solutions Architecture overlaps role_function. It is a skill that is also a role. The overlap is deliberate, so the kind of architect is distinguishable, but it must not be double-counted in Task 6.
4	Trend flags rest on 11 observation weeks after the censored one is dropped, with uneven gaps where partial scrapes were excluded. They indicate direction, not magnitude.
5	The taxonomy extension is unratiied. Until the team agrees it, my skill space is wider than the others', which would distort a similarity score in Task 8.

7 For the team

- Ratify the 34 added skills and the silicon_design category.** If the others extract against the original 59 and I extract against 93, Task 8 similarity measures the taxonomy, not the companies.
- Adopt a title match blocklist.** Any member matching short tokens against titles will hit the same r/R&D and c/c++ problems.
- 164 candidate terms are waiting**, ranked by frequency, in taxonomy_candidates_20260902.csv. Highest unmatched: infrastructure (98), factory (21), enterprise (20), compute (20).

Files

File	Rows	Contents
nvidia_skills_long_20260902.csv	1,793	The extracted-skills dataset. One row per posting-skill pair, with the matched term
nvidia_skill_features_20260902.csv	59	Per-skill feature table: counts, share, trend, emerging flag, seniority mix
nvidia_skill_by_week_20260902.csv	59 x 12	Skill by observation week, the input to Tasks 5 and 7
taxonomy_candidates_20260902.csv	164	Unmatched terms for the team's taxonomy review
extract_skills.py	-	The pipeline, re-runnable from a clean checkout

```
cd work/task-4/nvidia-nayab-khalid
python extract_skills.py
```

Declaration	No personal or sensitive data is present in any output. No new data was collected for this task; the input is the cleaned corpus committed in Task 3 and the shared taxonomy.
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